atom itself. That's why the quantum laws of physics deal with *probabilities* for where the electron is and not with its actual location (Figure 26.1).

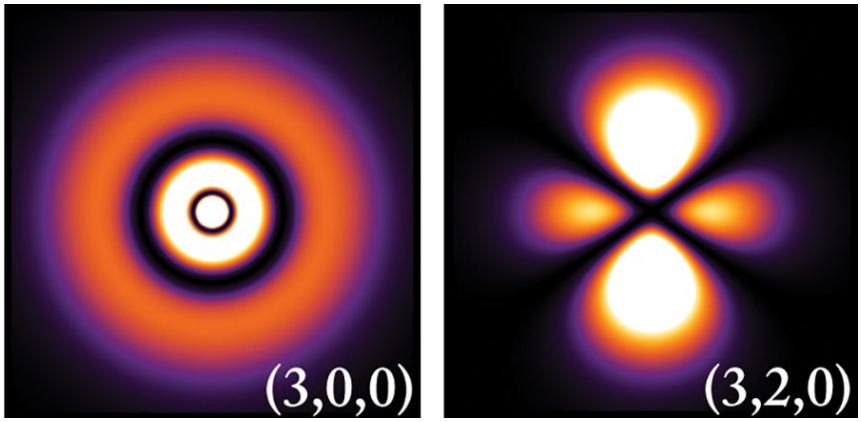


Fig. 26.1. Probability for electron's location inside two different hydrogen atoms. The probability is big in the white regions, smaller in the red, and very small in the black. The numbers (3,0,0) and (3,2,0) are the names of the two atoms' probability pictures.

When we use instruments to look at big things, we also see fluctuations, if our instruments are precise enough. But the fluctuations of big things are minuscule. In the LIGO gravitational wave detectors (Chapter 16), laser beams monitor the locations of hanging mirrors that weigh 40 kilograms (90 pounds).[45] Those locations fluctuate randomly, but by amounts far less than the size of an atom: *one ten-billionth of an atom's size*, in fact (Figure 26.2). Nevertheless, LIGO's laser beams will see those fluctuations a few years from now. (LIGO's design prevents those random fluctuations from getting in the way of measuring gravitational waves. My students and I helped make sure of this.)